Notes: Lian and Ma (QJE 2025)

Anatomy of Corporate Borrowing Constraints

Contents

1 Big picture: question, contribution, and "anatomy"

1.1 The core question

Macro-finance models often summarize corporate borrowing constraints using the liquidation value of pledged physical assets (the "collateral constraint" tradition). This paper asks: *Is that an empirically accurate description of how U.S. nonfinancial firms borrow and what constrains them on the margin?*

1.2 The core contribution

The paper provides a data-driven "anatomy" of corporate borrowing constraints for U.S. nonfinancial firms and argues that:

(i) **Most U.S. corporate debt is cash-flow based, not asset-collateral based.** Roughly 80% of debt by value is primarily supported by cash flows rather than by the liquidation value of pledged physical assets (Table I and Figure I).

(ii) **A standard and empirically relevant borrowing constraint is earnings-based.** A common constraint restricts debt as a multiple of operating earnings (EBITDA) and is implemented/enforced through earnings-based covenants (EBCs) in debt contracts (Section III; equations (2)–(3)).

(iii) **These constraints matter on the margin.** For firms where EBCs are relevant, borrowing and investment are sensitive to operating earnings (Table IV and Figure IV).

(iv) **Collateral value shocks are less central for these firms.** The sensitivity of borrowing to real estate values is limited in the main U.S. firm groups where cash-flow based lending dominates (Table VI), implying weaker collateral-damage and fire-sale amplification than standard collateral-constraint models would imply.

1.3 Why this matters conceptually

If the representative borrowing constraint is earnings-based rather than asset-collateral based, then:

- **Cash-flow shocks directly relax constraints** (earnings increase debt capacity mechanically).
- **Asset-price shocks are less "first order"** for constrained borrowing in the corporate sector (less collateral damage).
- **Fire-sale amplification is dampened** because the constraint is less directly tied to asset prices (Section V; equations (9)–(10)).

2 Debt taxonomy and measurement

2.1 Two types of debt: the creditor-parlance classification

The paper emphasizes a *contract/economic* distinction between:

TABLE I

COMPOSITION OF U.S. NONFINANCIAL CORPORATE DEBT

Panel A: Aggregate corporate debt share by type			
Category	Debt type	Share	
Asset-based lending (20%)	Mortgages	7%	
	Asset-based loans	13%	
Cash flow–based lending (80%)	Corporate bonds	57%	
	Cash flow–based loans	23%	
Panel B: Firm-level median share by group (Compustat)			
	Large firms	Rated firms	Small firms
Asset-based lending	11.1%	7.1%	53.5%
Cash flow–based lending	85.3%	90.5%	8.4%

Asset-based lending: debt backed by *specific, separable assets* with relatively well-defined liquidation value and enforceable claims over those assets (e.g., mortgages backed by real estate; asset-based loans backed by inventory/receivables).

Cash-flow based lending: debt backed predominantly by the *firm's going-concern value* and repayment capacity out of operating cash flows (e.g., most corporate bonds; most conventional corporate term loans and revolvers where the key underwriting object is cash flow).

2.2 Debt composition: what the data say (Table I and Figure I)

Aggregate composition (Table I, Panel A). U.S. nonfinancial corporate debt is approximately:

- **20% asset-based:** mortgages (7%) + asset-based loans (13%),
- **80% cash-flow based:** corporate bonds (57%) + cash-flow based loans (23%).

Firm-level composition (Table I, Panel B).

- **Large firms:** median cash-flow based share is very high (around 85%), while asset-based is low (around 11%).
- **Rated firms:** median cash-flow based share is even higher (around 90%); asset-based is very small (around 7%).
- **Small firms:** asset-based lending is much more important (median above 50%), consistent with weaker verifiable cash flows and limited access to broad credit markets.

Time-series prevalence (Figure I).

- Panel A: among large Compustat firms, the median share of cash-flow based lending remains persistently high (around 0.8–0.9) and slightly increases over time; the asset-based share is low (around 0.1–0.2) and declines.
- Panel B: the fraction of large firms with earnings-based covenants is substantial (roughly one-half to two-thirds) and varies over time.

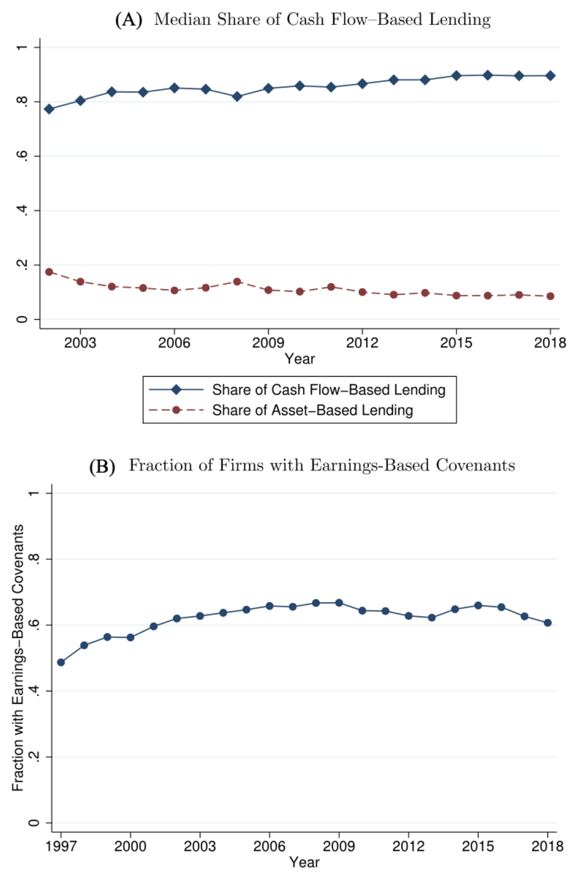

FIGURE I

Prevalence of Cash Flow–Based Lending and EBCs: Large Compustat Firms

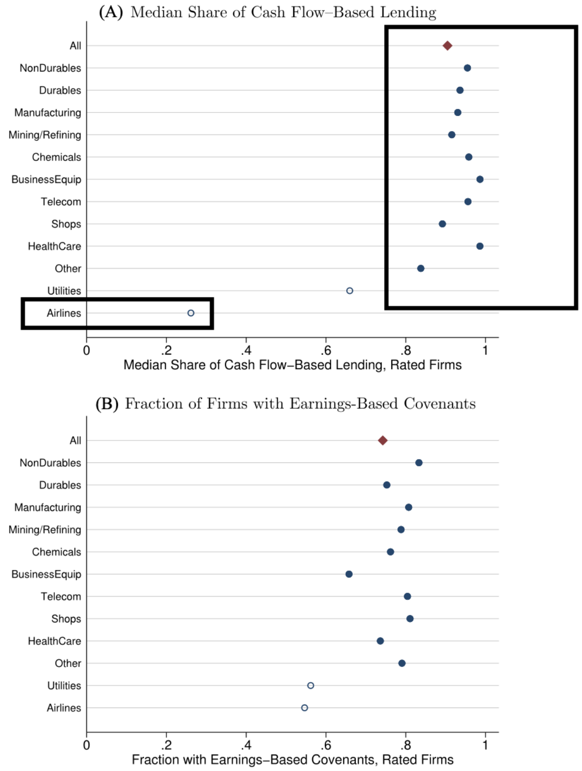

FIGURE II

Prevalence of Cash Flow–Based Lending and EBCs: Rated Firms by Industry

2.3 Industry heterogeneity (Figure II)

Figure II shows substantial cross-industry variation even among rated firms:

- Most industries exhibit very high median cash-flow based shares (near 0.9–1.0).
- Airlines (and to a lesser extent utilities) are outliers with much lower cash-flow based shares, consistent with historically asset-collateral intensive financing and/or specialized bankruptcy/liquidation considerations.
- The fraction of firms with EBCs is generally high where cash-flow based lending is prevalent, and lower in those outlier industries, linking cash-flow based lending to earnings-based constraints.

3 Model block 1: Properties of debt types (Table II)

3.1 Empirical model behind Table II

Table II estimates panel regressions that relate the amount of a given type of debt to the borrower's asset composition and to an estimated liquidation-value index:

$$\text{DebtType}_{i,t} = \alpha + \beta_1 \text{PPE}_{i,t} + \beta_2 \text{Inv}_{i,t} + \beta_3 \text{Rec}_{i,t} + \beta_4 \text{LiqVal}_{i,t} + \Gamma' Z_{i,t} + \varepsilon_{i,t}, \tag{1}$$

where (as described in the table notes) all variables are normalized by book assets, $Z_{i,t}$ includes controls such as firm size and cash holdings, and standard errors are clustered by firm and time.

TABLE II

Properties of Asset-Based Debt and Cash Flow–Based Debt

Panel A: Asset-based debt	All		Mortgage		Nonmortgage	
	(1)	(2)	(3)	(4)	(5)	(6)
Plant, property, and equipment	0.131***		0.032***		0.073***	
	(0.009)		(0.002)		(0.008)	
Inventory	0.040**		−0.000		0.051***	
	(0.016)		(0.002)		(0.013)	
Receivable	0.047***		−0.006***		0.058***	
	(0.012)		(0.001)		(0.012)	
Liquidation value		0.131***		0.026***		0.096***
		(0.016)		(0.003)		(0.015)
Observations	58,241	55,372	57,715	54,881	57,979	55,123
R^2	0.08	0.06	0.07	0.03	0.06	0.05

Panel B: Cash flow–based debt	All		Cash flow–based loans		Secured cash flow–based	
	(1)	(2)	(3)	(4)	(5)	(6)
Plant, property, and equipment	−0.105***		−0.061***		−0.066***	
	(0.012)		(0.008)		(0.008)	
Inventory	−0.230***		−0.065***		−0.073***	
	(0.016)		(0.011)		(0.008)	
Receivable	−0.330***		−0.090***		−0.112***	
	(0.023)		(0.010)		(0.011)	
Liquidation value		−0.358***		−0.134***		−0.154***
		(0.023)		(0.017)		(0.016)
Observations	58,227	55,361	57,309	54,467	57,495	54,652
R^2	0.07	0.06	0.05	0.05	0.04	0.04

3.2 Interpretation of Table II

Panel A: Asset-based debt is "about assets." Coefficients on asset components and on liquidation value are positive:

- **PPE, inventory, receivables** predict *more* asset-based debt.
- The **liquidation-value index** is strongly positively related to asset-based debt.

Economic meaning: lenders that write asset-based contracts scale lending with collateral value and collateralizable asset stocks.

Panel B: Cash-flow based debt is *not* scaled by these assets. Coefficients on the same asset measures and liquidation value are negative (or much smaller):

- Firms with more collateralizable assets rely less on cash-flow based debt *conditional on controls*.
- The liquidation-value index predicts *less* cash-flow based debt.

Economic meaning: cash-flow based contracts are underwritten on repayment capacity out of operating cash flows, so the marginal determinant is not liquidation value. This is consistent with the paper's "anatomy" claim: *for most large firms, the relevant borrowing constraint is not primarily collateral-value-based.*

4 Model block 2: Earnings-based borrowing constraints (EBCs)

4.1 EBC definitions (equations (1)–(2))

The paper formalizes EBCs as constraints that cap debt using operating earnings. Two canonical forms are:

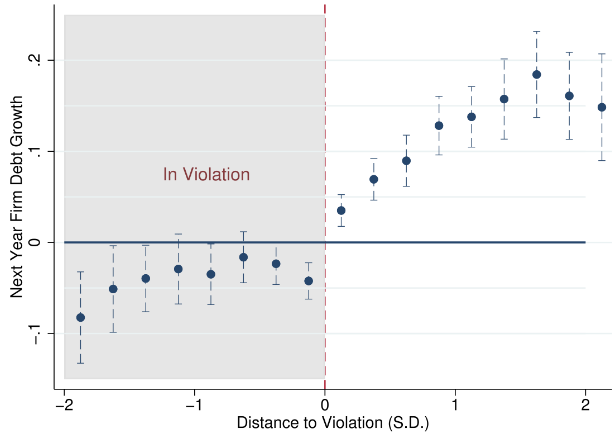

FIGURE III

Debt Growth and Earnings-Based Covenants

(i) Debt-to-earnings constraint.

$$b_t \leq \phi\, \pi_t, \tag{2}$$

where b_t is debt, π_t is annual operating earnings (e.g., EBITDA), and ϕ is the maximum allowed debt-to-earnings multiple.

(ii) Interest-coverage constraint. Let $r_t b_t$ be annual interest expenses. Then the constraint can be written as:

$$b_t \leq \theta\, \frac{\pi_t}{r_t}, \tag{3}$$

where θ is the maximum ratio of interest expense to earnings (equivalently the inverse of the minimum interest coverage ratio).

4.2 Contractual implementation: earnings-based covenants

EBCs are implemented in practice via *financial covenants* in debt contracts. The paper uses covenant data from DealScan (loans) and FISD (bonds) to measure:

- the prevalence of EBCs,
- the tightness of covenant thresholds (e.g., typical debt/EBITDA limits),
- the "distance to violation," a standardized measure of how close the borrower is to breaching the threshold.

4.3 Figure III: covenants constrain debt growth on the margin

Figure III plots next-year firm-level debt growth against the distance to violation of earnings-based covenants. The pattern is monotone:

- Stores/firms *in violation* (distance < 0) have very low or negative subsequent debt growth.
- As distance becomes positive (more slack), next-year debt growth rises.

TABLE IV

Debt Issuance and Investment Activities: Large Firms with Earnings-Based Covenants

Panel A: Baseline results	Net debt issuance		ΔLT book debt		CAPX		R&D	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EBITDA	0.281***	0.283***	0.409***	0.379***	0.111***	0.081***	0.035***	0.028**
	(0.041)	(0.040)	(0.051)	(0.045)	(0.015)	(0.017)	(0.011)	(0.013)
OCF		−0.004		0.057		0.057***		0.011
		(0.040)		(0.044)		(0.012)		(0.012)
Q	0.011**	0.011**	0.005	0.005	0.009***	0.008***	0.004***	0.004***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.001)	(0.001)	(0.001)	(0.001)
Past 12m stock ret	−0.003	−0.003	−0.003	−0.002	0.005**	0.006**	−0.002***	−0.002***
	(0.004)	(0.004)	(0.005)	(0.005)	(0.002)	(0.002)	(0.001)	(0.001)
L.Cash holding	0.015	0.014	0.058*	0.060**	−0.002	0.001	0.013	0.013
	(0.028)	(0.029)	(0.030)	(0.031)	(0.008)	(0.009)	(0.009)	(0.009)
Controls					Yes			
Fixed effects					Firm, Year			
Observations	20,675	20,675	22,072	22,059	22,107	22,107	11,308	11,306
R^2	0.11	0.11	0.12	0.12	0.14	0.15	0.11	0.11

Panel B: State dependence	Net debt issuance				
	(1)	(2)	(3)	(4)	(5)
EBITDA	0.166***	0.267***	0.266***	0.172***	0.171***
	(0.063)	(0.043)	(0.041)	(0.057)	(0.059)
Tight	−0.025***			−0.025***	−0.025***
	(0.009)			(0.009)	(0.009)
Tight × EBITDA	0.136**			0.139**	0.140**
	(0.056)			(0.054)	(0.056)
Output gap (demeaned) × EBITDA		−3.397**		−3.351**	
		(1.398)		(1.432)	
Unemployment rate (demeaned) × EBITDA			0.045***		0.044***
			(0.017)		(0.017)
Controls			Yes		
Fixed effects			Firm, year		
Observations	19,352	19,352	19,352	19,352	19,352
R^2	0.10	0.10	0.10	0.10	0.10

This is an "on-the-margin" diagnostic: the covenant threshold is a *kink/constraint boundary* shaping the feasible debt-growth set.

5 Model block 3: Borrowing sensitivity to operating earnings (Section IV)

5.1 Baseline panel specification (equation (3))

To quantify how operating earnings map into borrowing and investment, the paper estimates:

$$Y_{it} = \alpha_i + \eta_t + \beta\, EBITDA_{it} + X_{it}'\gamma + \varepsilon_{it}, \tag{4}$$

where Y_{it} is typically net debt issuance (and also alternative debt/investment outcomes), and the specification includes firm fixed effects α_i and year fixed effects η_t.

Interpretation of β. Within a firm, holding constant a rich set of controls (including internal funds measures), β measures how much additional external borrowing is associated with higher operating earnings. Under EBCs, the mechanism is direct: a higher $EBITDA_{it}$ relaxes the constraint in (2)–(3) and mechanically increases debt capacity.

5.2 State dependence: interactions (Table IV, Panel B)

The paper allows the sensitivity to vary with a state variable S_{it}:

$$Y_{it} = \alpha_i + \eta_t + \beta\, EBITDA_{it} + \kappa S_{it} + \varphi\,(EBITDA_{it} \times S_{it}) + X_{it}'\gamma + \varepsilon_{it}. \tag{5}$$

Examples of S_{it} in Table IV:

- a **Tight** indicator for covenants being close to binding,
- macro conditions such as demeaned **output gap** or **unemployment rate**.

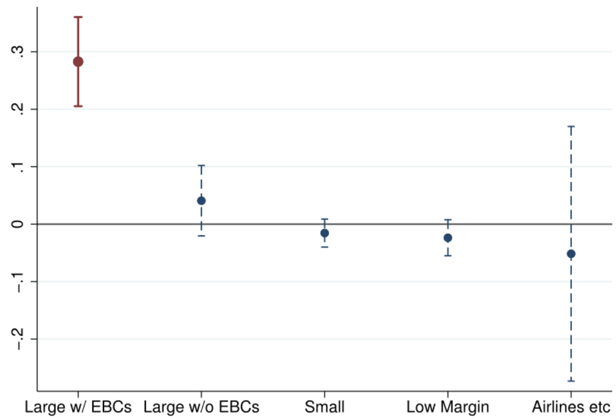

FIGURE IV
Borrowing Sensitivity to Operating Earnings by Firm Group

5.3 Interpretation of Table IV

Table IV focuses on **large U.S. nonfinancial firms with earnings-based covenants**, where the EBC mechanism is most relevant.

Panel A: baseline results. Key messages:

- **Borrowing:** $EBITDA_{it}$ has a large positive and highly significant coefficient in net debt issuance regressions. Economically: higher operating earnings relax EBCs and permit more borrowing.
- **Investment:** $EBITDA_{it}$ also predicts higher CAPX and R&D, consistent with borrowing capacity transmitting into real activity.
- **Internal funds controls:** the paper controls for operating cash flow and other balance-sheet measures; the EBITDA coefficient remains strong, consistent with a *constraint channel* rather than purely "more internal cash".

Panel B: state dependence. Key messages:

- When covenants are **tight**, the intercept level of borrowing is lower and the responsiveness to EBITDA changes.
- In **worse macro conditions** (lower output gap / higher unemployment), the EBITDA sensitivity increases, consistent with constraints binding more strongly in downturns.

5.4 Figure IV: heterogeneity across firm groups

Figure IV summarizes the estimated β from (4) across firm groups:

- **Large firms with EBCs:** strong positive sensitivity of borrowing to EBITDA (statistically precise).

- **Large firms without EBCs:** much weaker sensitivity (often close to zero), consistent with being far from constraint boundaries.
- **Small firms / low-margin firms / airlines & utilities:** little or no positive sensitivity, consistent with lower prevalence of cash-flow based lending and EBCs; borrowing constraints there are more plausibly asset-based or otherwise structured.

6 Model block 4: Identification via an accounting natural experiment (SFAS 123(r), Table V)

6.1 Why this helps identification

A generic concern is that $EBITDA_{it}$ could co-move with unobserved demand for credit, investment opportunities, or other fundamentals. To isolate a plausibly exogenous component of EBITDA, the paper uses a change in accounting rules for stock-based compensation.

6.2 The accounting change (intuition)

SFAS 123(r) required firms to expense stock-option compensation in operating expenses. This reduces reported operating earnings (EBITDA) *without necessarily changing real cash flows or real productivity.* If borrowing constraints are earnings-based, then a mechanical reduction in EBITDA can tighten covenants and reduce borrowing.

6.3 Cross-sectional IV specification (equation (4))

The paper estimates a cross-sectional IV regression for fiscal year 2006:

$$Y_i^{2006} = \alpha + \beta \, \widehat{EBITDA}_i^{2006} + X_i'\gamma + \varepsilon_i, \tag{6}$$

where $EBITDA_i^{2006}$ is instrumented by pre-adoption option compensation expenses (average over 2002–2004).

6.4 Interpretation of Table V

Panel A (first stage). Firms with higher pre-2004 option compensation expenses experience larger declines in reported EBITDA after adoption, generating a strong first stage.

Panel B (IV results). The key economic reading is group-specific:

- **Large firms with EBCs:** instrumented EBITDA has a positive effect on net debt issuance and CAPX. Thus, an exogenous reduction in EBITDA (via accounting) reduces borrowing and investment, consistent with an EBC channel.
- **Large firms without EBCs:** effects differ and can be weaker or even negative, consistent with not being constrained by earnings-based limits (or adjusting differently because they are far from covenant boundaries).

TABLE V

CHANGES IN EBITDA: ACCOUNTING NATURAL EXPERIMENT

Panel A: First stage

	EBITDA_i^{06} Large w/ EBCs (1)	Large w/o EBCs (2)	Small (3)
Avg opt comp expense 2002-2004	-0.806^{***}	-1.073^{***}	-0.623^{***}
	(0.169)	(0.231)	(0.240)
Controls		Yes	
Observations	668	434	843

Panel B: IV

	Net debt issuance			CAPX		
	Large w/ EBCs (1)	Large w/o EBCs (2)	Small (3)	Large w/ EBCs (4)	Large w/o EBCs (5)	Small (6)
$\widehat{EBITDA}_i^{06}$	0.964^{**}	-0.532^{**}	0.311	0.500^{***}	0.145	-0.093
	(0.398)	(0.235)	(0.203)	(0.125)	(0.150)	(0.112)
Controls			Yes			
First-stage F	19.74	18.47	5.95	18.80	27.27	7.85
Observations	668	434	843	665	431	833

TABLE VI

CORPORATE BORROWING AND REAL ESTATE VALUE

	Net debt issuance		ΔAsset-based		ΔCash flow–based		CAPX	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
RE (Method 1)	0.034^{**}		0.027		0.003		0.042^{***}	
	(0.017)		(0.017)		(0.025)		(0.010)	
RE (Method 2)		0.016		0.053^{***}		-0.002		-0.014
		(0.035)		(0.016)		(0.051)		(0.014)
EBITDA	0.309^{***}	0.163^{***}	0.151^{***}	0.095^{***}	0.325^{***}	0.158^{***}	0.066^{***}	0.057^{***}
	(0.100)	(0.040)	(0.043)	(0.036)	(0.105)	(0.043)	(0.013)	(0.015)
OCF	-0.153^{***}	-0.142^{***}	-0.121^{**}	-0.188^{***}	-0.096	-0.025	-0.011	-0.012
	(0.053)	(0.054)	(0.049)	(0.032)	(0.081)	(0.062)	(0.010)	(0.009)
Q	0.025^{***}	0.016^{***}	-0.002	0.005^{*}	0.019^{***}	0.010^{*}	0.006^{***}	0.007^{***}
	(0.008)	(0.005)	(0.002)	(0.003)	(0.007)	(0.006)	(0.002)	(0.001)
Past 12m stock ret	-0.008^{**}	-0.006	-0.005	-0.004	-0.007	-0.003	-0.000	0.003^{**}
	(0.004)	(0.004)	(0.004)	(0.003)	(0.005)	(0.004)	(0.001)	(0.001)
L.Cash holding	-0.109^{***}	-0.065^{**}	0.003	-0.008	-0.097^{***}	-0.071^{*}	0.017	0.001
	(0.031)	(0.032)	(0.022)	(0.022)	(0.028)	(0.038)	(0.012)	(0.006)
Controls				Yes				
Fixed effects				Firm, Year				
Observations	5,751	5,776	5,751	5,776	5,751	5,776	5,749	5,771
R^2	0.11	0.09	0.03	0.03	0.06	0.06	0.11	0.10

- **Small firms:** effects are imprecise, consistent with different financing technology (more asset-based, less covenant-driven).

Why this is compelling. The accounting instrument shifts the measured earnings variable that appears in covenants. Thus the IV estimate supports the interpretation that $EBITDA$ matters for borrowing *because it is a contractual input into constraints*, not merely because it proxies for internal funds.

7 Model block 5: Collateral value (real estate) and borrowing (Table VI)

7.1 Panel regression on real estate value (equation (5))

To evaluate the classic collateral-value channel, the paper estimates:

$$Y_{it} = \alpha_i + \eta_t + \beta\, RE_{it} + X_{it}'\gamma + \varepsilon_{it}, \tag{7}$$

where RE_{it} is the market value of real estate holdings (computed using two measurement methods), and Y_{it} includes:

- net debt issuance,
- changes in asset-based debt outstanding,
- changes in cash-flow based debt outstanding,
- capital expenditures (CAPX).

7.2 Interpretation of Table VI

The table supports a sharp implication of the "anatomy":

- **Real estate value has limited predictive power for overall borrowing and investment** once controls and fixed effects are included.
- Any sensitivity to RE_{it} is concentrated, if at all, in **asset-based debt** (the debt type that is mechanically collateral-linked).
- **Cash-flow based debt does not respond to real estate value** in the way collateral-constraint models would imply.

Thus, for the dominant portion of debt (cash-flow based), the "collateral value" state variable is not the key driver of debt capacity.

8 Model block 6: Great Recession collateral-damage test (equation (8))

To directly evaluate a collateral-damage channel in the Great Recession, the paper estimates the following cross-sectional specification:

$$\Delta Y_i^{07\text{–}09} = \alpha + \lambda\, \Delta RE_{i,06}^{07\text{–}09} + \eta\, RE_i^{06} + \phi\, \Delta P_i^{07\text{–}09} + \beta\, \Delta EBITDA_i^{07\text{–}09} + X_i'\gamma + u_i, \tag{8}$$

where:

- $\Delta Y_i^{07\text{–}09}$ is the change in debt issuance or investment from 2007 to 2009,
- $\Delta RE_{i,06}^{07\text{–}09}$ is the value loss/gain of precrisis real estate holdings due to property-price changes (scaled by assets),

- $\Delta P_i^{07\text{–}09}$ controls for local property-price changes that can affect demand through local household balance-sheet channels,
- $\Delta EBITDA_i^{07\text{–}09}$ captures cash-flow deterioration,
- RE_i^{06} controls for precrisis real estate exposure levels.

Interpretation. This specification separates two channels:

- **Collateral damage:** through λ on ΔRE,
- **Earnings channel:** through β on $\Delta EBITDA$.

The paper finds that the earnings channel can account for a meaningful share of the contraction in borrowing and investment for EBC-relevant firms, while collateral damage through real estate is limited for major U.S. nonfinancial firms.

9 Model block 7: Macro modeling implications and the Kiyotaki–Moore comparison (Section V)

9.1 Two alternative borrowing constraints in a standard GE environment

To illustrate why the *form* of constraints matters for amplification, the paper compares:

(i) Traditional collateral constraint (equation (6)).

$$b_t \leq \frac{1-\delta}{R} q_{t+1} k_t, \tag{9}$$

where b_t is borrowing, k_t is capital, q_{t+1} is the liquidation value of capital next period, R is the gross interest rate, and δ is depreciation.

(ii) Earnings-based constraint (equation (7)).

$$b_t \leq \frac{\theta}{R} a_t k_t, \tag{10}$$

where $a_t k_t$ is the operating earnings object in the simple production environment and θ governs the tightness of the EBC.

9.2 Intuition: why amplification differs

- Under (9), an adverse shock that reduces asset prices q_{t+1} tightens borrowing capacity directly, forcing deleveraging and potentially triggering fire sales that further depress q_{t+1} (feedback loop).
- Under (10), borrowing capacity is tied to operating earnings ($a_t k_t$). There is no *direct* feedback from asset prices into constraints, so the "collateral damage" mechanism is muted and fire-sale amplification is dampened.

9.3 Mixed economy calibration (mapping to the paper's debt shares)

Because only a minority of debt is asset-based in the U.S. corporate sector, the paper argues that a realistic macro specification should incorporate a large EBC component. In a mixed economy where most firms/debt face (10), the aggregate amplification is much smaller than in an economy where all firms face (9).

10 Summary

1. **Debt type matters for the right macro friction.** "Borrowing constraint = collateral value" is not an accurate reduced form for most large U.S. nonfinancial firms.
2. **Earnings-based constraints are empirically central.** Contracting via EBCs (debt/EBITDA, interest coverage) is widespread and binds on the margin.
3. **Operating earnings relax constraints.** Borrowing and investment respond strongly to EBITDA for EBC-relevant firms, and this relationship survives an accounting-based IV design.
4. **Collateral damage is limited for the dominant debt component.** Real estate values do not strongly drive borrowing for the main firm groups, implying weaker fire-sale amplification in the corporate sector than standard collateral-constraint models assume (in the U.S. institutional environment).

11 Conclusion and intuition

11.1 Coherent summary: what the paper concludes

This paper revisits a foundational modeling choice in macro–finance: *what object constrains corporate borrowing on the margin.* A large class of models ties corporate debt capacity to the *liquidation value of pledgeable physical assets* (a "collateral constraint"). Using detailed debt-type data and covenant data for U.S. nonfinancial firms, the paper concludes that this is not the empirically dominant borrowing technology for the U.S. corporate sector.

The paper's conclusions can be summarized as follows:

(i) **Corporate borrowing constraints are mostly cash-flow based in the U.S.** For U.S. nonfinancial firms, about **20%** of debt by value is *asset-based* (mortgages and asset-based loans), whereas about **80%** is *cash-flow based* (corporate bonds and standard corporate loans). Hence, for most large firms, debt capacity is not primarily determined by the resale/liquidation value of specific pledged assets.

(ii) **A standard and empirically relevant constraint is earnings-based.** In cash-flow based lending, a common constraint restricts total debt as a (contractual) function of operating earnings (e.g., debt-to-EBITDA or interest-coverage style limits). The paper calls these *earnings-based borrowing constraints* (EBCs) and shows they are prevalent and often binding.

(iii) **Earnings-based constraints shape outcomes on the margin.** When EBCs are present (especially among large firms), borrowing and investment respond strongly to operating earnings: higher operating earnings mechanically relax constraints and expand debt capacity, while covenant tightness/kinks reduce subsequent debt growth and can propagate shocks through debt issuance and investment.

(iv) **Collateral-value channels are less central for the dominant debt component.** For the large-firm segment where cash-flow based borrowing dominates, the link between borrowing/investment and collateral value (e.g., real estate values) is limited relative to what collateral-constraint models imply. This suggests that collateral-damage and fire-sale amplification mechanisms may be *muted* in the U.S. corporate sector (though they may remain important in settings where asset-based debt is dominant).

(v) **Modeling should adapt to the institutional enforcement environment.** The paper emphasizes that which constraint is realistic depends on legal/institutional features of enforcement and bankruptcy. Cross-country differences in debt enforcement imply that macro–finance mechanisms will not apply uniformly across economies.

11.2 Economic intuition: why earnings-based constraints dominate in the U.S.

The central intuition is that *contract form is shaped by what is enforceable and observable.*

1) Enforcement shapes what creditors can reliably claim. In many theoretical setups, creditors can easily seize physical assets, and cash flows are hard to verify. In that world, it is natural that debt capacity is tied to pledgeable liquidation value. In contrast, in the U.S. institutional environment, creditors may *not* be able to seize and liquidate broad sets of physical assets in a frictionless way (e.g., because bankruptcy procedures such as the automatic stay limit immediate seizure), while firms may have *highly verifiable* operating earnings.

2) Verifiability and contract simplicity favor EBITDA-multiple rules. It is operationally straightforward to write (and monitor) covenants like

$$\text{Debt} \le \phi \cdot \text{EBITDA} \quad \text{or} \quad \frac{\text{EBITDA}}{\text{Interest}} \ge \text{Threshold},$$

because the inputs are accounting objects that can be verified repeatedly. By contrast, writing a "total debt $\le$ liquidation value of pledgeable assets" rule would require continual valuation of heterogeneous assets and clear enforcement of liquidation rights, which is not the comparative advantage of cash-flow lenders and is often not feasible at scale.

3) Hence, the marginal constraint reacts to cash-flow shocks, not asset-price shocks. If the binding constraint is earnings-based, then *operating earnings* are the state variable that moves debt capacity. A fall in EBITDA tightens constraints immediately and reduces borrowing/investment even if asset prices are stable. Conversely, an asset price decline (e.g., real estate) has weaker first-order effects on total borrowing capacity for firms whose debt is not primarily collateralized by those assets.

11.3 Implications for macro–finance and empirical work

- **Macro modeling:** For U.S. nonfinancial corporates, using a representative collateral constraint for all firms may overstate collateral-damage amplification and understate the direct cash-flow channel through covenant-based debt capacity. A more realistic "workhorse" friction for this sector is an earnings-based borrowing constraint.

- **Heterogeneity:** Collateral constraints still matter where asset-based lending is prevalent (e.g., smaller firms or sectors with more standardized pledgeable assets), so a realistic aggregate model is naturally a *mixed* economy with both constraint types.

- **Interpretation of "net worth channel":** The general lesson that external finance is costly and internal funds matter remains valid. What the paper adds is that the *form* of the constraint—earnings-based versus collateral-value-based—matters for which shocks propagate and which amplification mechanisms are empirically relevant.

11.4 Takeaway

For U.S. nonfinancial firms, borrowing is predominantly cash-flow based and often constrained by earnings-based covenants (e.g., debt/EBITDA), so operating earnings shocks directly move debt capacity while collateral-value channels and fire-sale amplification are comparatively less central for the dominant debt component.